probability sample exam 1

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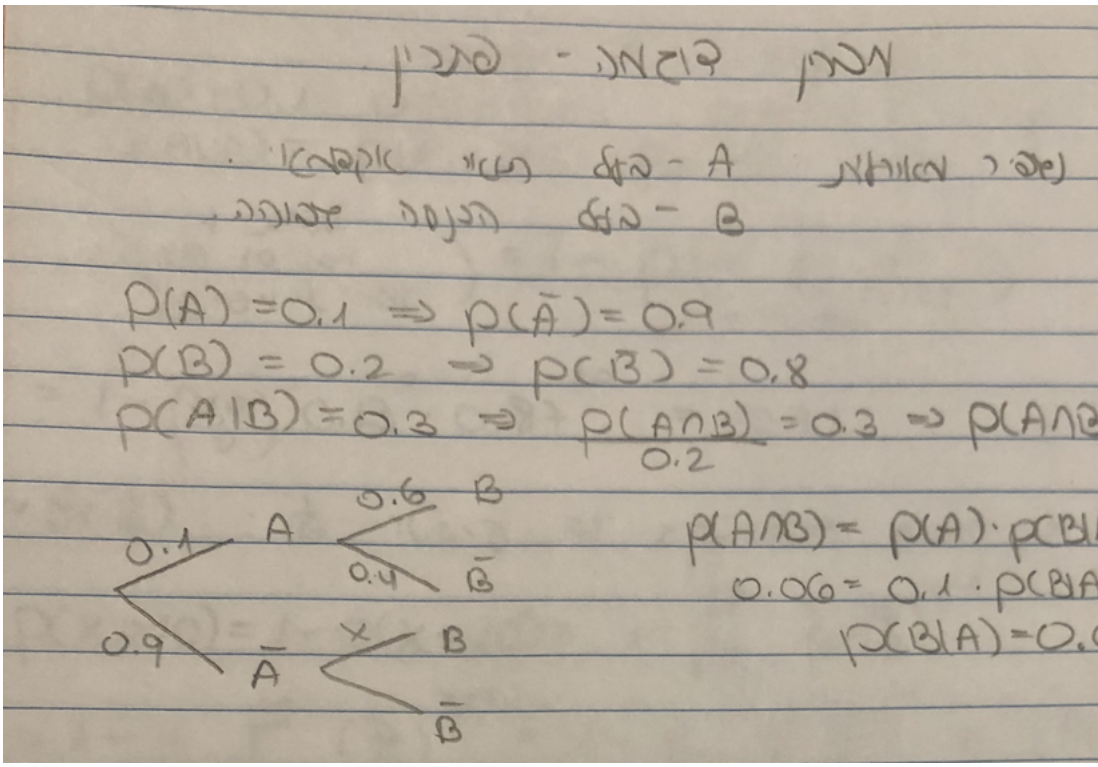
Question 1. in a club 10% have degrees and 20% have high income, 30% of the ones with high income have a degree. picking a person from the club at random that has a degree, what are the odds they also have high income?

Answer 1.0.1.



Question 2. the probability of rain on holiday 1 is 1/4, on holiday 2 its 1/3, those events are independent, what is the probability of rain on holiday 1 or holiday 2?

Answer 2.0.1. use the inclusion-exclusion principle, its 1/4+1/3-1/4*1/3

Question 3. 2 persons are elected, both have equal probabiblity of being voted for by any given voter

Subquestion 3.1. what is the probability that of 5 people more than 75% vote for the second person

Answer 3.1.1. probabilities of 4 and 5 added, each found using the formula Bernoulli distribution

Subquestion 3.2. what is the probability that the vote of the fifth person is different from the rest

Answer 3.2.1. $\left(\frac{1}{2}\right)^4 \cdot \frac{1}{2} + \left(\frac{1}{2}\right)^4 \cdot \frac{1}{2}$

Question 4. 10% are rich, 5% are famous, 2% are rich and famous, what is the probability that of 10 people atleast one is famous or rich

Answer 4.0.1. A - rich
B - famous

$$\begin{aligned}
P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\
&= 0.1 + 0.05 - 0.02 \\
&= 0.13 \\
P\left(\overline{A} \cap \overline{B}\right) &= 1 - P(A \cup B) \\
&= 1 - 0.13 = 0.87 \\
\text{answer} &= 1 - 0.87^{10}
\end{aligned}$$

Question 5. we throw a pair of dies until the same result is received in both

Subquestion 5.1. let X be the total number of throws, what is the distribution of X?

Answer 5.1.1. here X is a continuous random variable
it has a , $X \sim G\left(\frac{1}{6}\right)$

Subquestion 5.2. what is the probability that the pair of dies is thrown more than 10 times?

Answer 5.2.1.

$$P(X > 10) = 1 - \sum_{i=1}^{10} \left(\frac{5}{6}\right)^{i-1} \frac{1}{6}$$

or simply

$$\left(\frac{5}{6}\right)^{10}$$

Question 6. in a class theres 7 students, 4 girls and 3 boys, we pick 2 students, we denote by X the number of boys picked, what is the expected value and variance of X?

Answer 6.0.1. we use the hypergeometric distribution

$$X = \begin{cases} 0 & \frac{\binom{3}{0}\binom{7-3}{2-0}}{\binom{7}{2}} = \frac{2}{7} \\ 1 & \frac{\binom{3}{1}\binom{7-3}{2-1}}{\binom{7}{2}} = \frac{4}{7} \\ 2 & \frac{\binom{3}{2}\binom{7-3}{2-2}}{\binom{7}{2}} = \frac{1}{7} \end{cases}$$

the expected value is $0 \cdot \frac{2}{7} + 1 \cdot \frac{4}{7} + 2 \cdot \frac{1}{7} = \frac{6}{7}$
the variance is

$$\mathrm{Var}(X) = E\left(\left(X - \frac{6}{7}\right)^2\right) = \left(0 - \frac{6}{7}\right)^2 \cdot \frac{2}{7} + \left(1 - \frac{6}{7}\right)^2 \cdot \frac{4}{7} + \left(2 - \frac{6}{7}\right)^2 \cdot \frac{1}{7}$$

Question 7. the number of people that arrive at a bus station in fifteen minutes is a random variable with a poisson distribution with parameter $\lambda = 10$

Subquestion 7.1. what is the probability that from 10:00 to 10:15 7 people arrive at the bus station?

Answer 7.1.1.

$$P(X = 7) = \frac{10^7}{7!}e^{-10}$$

Subquestion 7.2. what is the probability that from 20:15 to 20:30 at most 3 people arrive at the bus station?

Answer 7.2.1. $\frac{10^0}{0!}e^{-10} + \frac{10^1}{1!}e^{-10} + \frac{10^2}{2!}e^{-10} + \frac{10^3}{3!}e^{-10}$

Question 8. an investment portfolio consists of 10 independent stocks. the chance that a stock will go up on a certain day is 40%. the owner of the portfolio will sell it on the day when at least 8 shares (out of the 10) will rise. what is the expected number of days until the owner of the portfolio sells it? in the solution, a random variable must be defined and its distribution found.

Answer 8.0.1. we find p using the formula for the Bernoulli distribution
let X be the number of days when 8 out of 10 stocks rise
$X \sim G(p)$

Question 9. a balanced spinner is spun twice with the numbers 1,2,3,4 written on its ends. let X be the higher number of the two spins, and let Y be the lower number of the two spins.

Subquestion 9.1. find the joint probability distribution of X and Y

note

Answer 9.1.1. 2/16 because if we get for example 1 and then 2 or 2 and then 1

X \ Y	1	2	3	4	P(X=x)
1	1/16	0	0	0	1/16
2	2/16	1/16	0	0	3/16
3	2/16	2/16	1/16	0	5/16
4	2/16	2/16	2/16	1/16	7/16
P(Y=y)	7/16	5/16	3/16	1/16	1